

ASSESSMENT GUIDE

Unit: ITEC610 Python fundamentals for data science, Semester 2, 2024

Assessment number (2)

Assessment Artefact: Python Codes and Comments

Weighting [30%]

Why this assessment?

- The purpose is to assess students' use of Python basics, e.g., lists and functions, as well as Python data science libraries NumPy and Pandas to explore data.

What are the types of employability skills that I will acquire upon completion of this assessment?

Skill Type	
Developed critical and analytical thinking	☒
Developed ability to solve complex problems	☐
Developed ability to work effectively with others	☐
Developed confidence to learn independently	☒
Developed written communication skills	☒
Developed spoken communication skills	☐
Developed knowledge in the field study	☐
Developed work-related knowledge and skills	☒

Assessment Overview:

Purpose, as written in the EUO

Due date:	5pm on Friday of Week 09 (04 October 2024)
Weighting:	30%
Length and/or format:	Python codes, comments and markdown in Colab notebook
Learning outcomes assessed	LO2
Graduate attributes assessed	GA5, GA10
How to submit:	task submission – via Canvas
Return of assignment:	Via Canvas within 2 weeks of submission
Assessment criteria:	Rubric: see end of document

Context

Data processing lab practical with Python basics and Python libraries NumPy and Pandas

Coronavirus disease 2019 (**COVID-19**) is a contagious disease caused by severe acute respiratory syndrome **coronavirus 2** (SARS-CoV-2). The first case was identified in Wuhan, China, in December 2019. The disease has since spread worldwide, leading to an ongoing pandemic.

Monitoring the spread of the coronavirus disease is expected to monitor epidemiological trends, rapidly detect new cases, and based on this information, provide epidemiological information to conduct risk assessment and guide disease preparedness.

In this practical you will have the chance to do initial exploratory about a COVID-19 dataset with learned skills of Python basics and Python data science libraries NumPy and Pandas.

Instructions

You will be given a COVID-19 dataset read in from python code, and are required to do below tasks:

1. **(4.5 marks, 1.5 marks for each)** Good structure of Python Jupyter Notebook
 - a. Containing title cells, subtitle cells.
 - b. Python codes are reasonable separated into groups (code cells) with functionalities.
 - c. Containing meaningful comments and sensible variable and function names.
2. **(10 marks, 2 marks each))** download csv data with pandas with below code:
 - import pandas as pd
 - ```
deaths_df = pd.read_csv('https://github.com/CSSEGISandData/COVID-19/blob/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_deaths_global.csv')
```
  - Identify and handle missing values in the dataset.
  - Remove duplicate entries if any.
  - Convert the date column to a consistent date format (e.g., YYYY-MM-DD)
  - Save the cleaned dataset to a new CSV file.
3. **(3 marks)** Display first 5 rows of the loaded data (1 mark) and do a short summary about the data (2 marks)
4. **(2.5 marks)** Calculate the mean and median of the daily cases.
5. **(3 marks)** Get daily death cases worldwide (hint: summarizing daily death cases over all countries.)
6. **(3 marks)** Get daily increasement of deaths cases via defining a function (hint: use the death cases of today minus the death cases of yesterday from the data obtained in task 5.)
7. **(4 marks)** Visualize the data obtained in task 5 with library matplotlib

## Structure

Prepare a Colab notebook for this assignment. The structure of the Colab notebook should alternate texts and python codes and cover topics listed the in specific tasks above. One template could be found in any week's workshop resources in LEO.

## How do I submit?

1. Prepare your Colab notebook (.ipynb) with your solutions, code, and explanations.
2. Create a PDF file that contains screenshots of your solutions and the corresponding output results. Ensure that your PDF is clear and legible.
3. Include any supporting documents, if necessary, to enhance the clarity and completeness of your submission.
4. Navigate to the "Assessment 2" link via the Canvas assessment tile.
5. Upload your Colab notebook, PDF file, and any supporting documents to the designated submission area.
6. Double-check your submission to ensure that all files are attached correctly.
7. Submit your assessment within the specified deadline.

**Note that:** The code will be compared to other students' submission in Turnitin / ChatGPT to make sure the submission satisfies.

academic integrity.

### Submission checklist

|                                                                                                                            |                          |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------|
| I have formatted my report as per the specifications                                                                       | <input type="checkbox"/> |
| I have checked my Turnitin report and taken appropriate actions to ensure that the submission satisfies academic integrity | <input type="checkbox"/> |
| I have actioned feedback advice provided to me from labs feedback (if applicable)                                          | <input type="checkbox"/> |
| I have submitted my work before the due date/time                                                                          | <input type="checkbox"/> |
| I have submitted feed forward template along with my assignment submission                                                 | <input type="checkbox"/> |

### Feed Forward Template (example)

A template for students to use and act on feedback and provide recommendations for improvement.

#### Note

This is a task for any instance of follow-on assignment (assessment 2 and 3). This must be submitted as the first page of the follow-on assignment (assessment 2 and 3) to ensure you acted on the feedback provided to you in the previous assignment (this is not counted as part of the assessment word count).

#### How did you act on the feedback?

Feedback is an important component of learning. Please consider the feedback you received in your last assignment and provide a response on how you acted on, or intend to act upon, that feedback, and how it has informed the current assignment task. Submit this sheet along with your assignment.

| Questions                                                                                                                                                                                                | Your learning from the previous assignment feedback                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How have you acted on the feedback from previous assignment to improve your work in this assignment?                                                                                                     | (e.g. based on my previous feedback, I made sure that I supported my discussion, position, ideas, concepts with peer reviewed journal references in this assignment)                                                                                                 |
| What is your expectation around the type of feedback that enhances your learning?                                                                                                                        | (e.g. I want to know where I made a mistake and how I can correct them and not make the same mistake again i.e. I want specific feedback that will help me to improve my learning and performance in the next assignment)                                            |
| Did you have any difficulty understanding or acting on previous feedback? Please be as specific as possible so that you can gain further feedback/clarify anything you do not understand in the feedback | (e.g. feedback provided in my previous assignment was very generic I did not know how to improve my work. So, I would like the teacher to explain more on xxxx aspects of the feedback or I would like an opportunity to have a dialogue to understand the feedback) |

## Some Helpful Websites and Resources

LEO weekly materials

Google CoLab environment <https://colab.research.google.com/notebooks/>

Python official website <https://www.python.org/>

Useful python packages:

NumPy <https://numpy.org/>

Pandas <https://pandas.pydata.org/>

## Who can help me?

Academic skills Unit (ASU)

LiC:- Dr. Walayat Hussain

## I'm having problems

**Special Consideration:** This [form](#) is used by students to apply for **Special Consideration** for assessable work in studies at Australian Catholic University. Approval of such applications will only be granted to students who are legitimately disadvantaged in their assessment due to exceptional and unforeseen **circumstances** beyond their control.

## Referencing

N/A

## Criteria

The full criteria is compiled in a rubric, which can be found on the following page/s.

## Rubric for Assessment 2

| Relevant LO/GAs                                                                                       | Criterion (related to a single GA from the related LO – one GA per criterion)                                                                                        | Does not meet expectations                                                                                                                                                                                                         | Meets expectations                                                                                                                                                                                                               |                                                                                                                                                                                                                                      | Exceeds expectations                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                       |                                                                                                                                                                      | NN (0-49)                                                                                                                                                                                                                          | PA (50-64)                                                                                                                                                                                                                       | CR (65-74)                                                                                                                                                                                                                           | DI (75-84)                                                                                                                                                                                                                                        | HD (85-100)                                                                                                                                                                                                                                                                                 |
| <b>GA5</b><br><b>LO2</b><br><b>Weight=15 marks</b><br><b>TL=3</b><br><b>Learning stage = I and D</b>  | Demonstrate skills of correct understanding and use of Python basics including data structures e.g., lists, control flow e.g. if-else and loops as well as functions | Demonstrates an insufficient understanding of Python basics, including data structures, control flow, and functions. Python code is disorganized, and comments and variable/function names are unclear.                            | Demonstrates a basic understanding of Python basics, including data structures, control flow, and functions. Python code lacks organization, and comments and variable/function names are not meaningful.                        | Demonstrates a basic understanding of Python basics, including data structures, control flow, and functions. Python code lacks organization, and comments and variable/function names are not meaningful.                            | Demonstrates a satisfactory understanding of Python basics, including data structures, control flow, and functions. Python code is somewhat organized, but comments and variable/function names lack clarity at times.                            | Demonstrates a proficient understanding of Python basics, including data structures, control flow, and functions. Python code is organized, and comments and variable/function names are mostly meaningful.                                                                                 |
| <b>GA10</b><br><b>LO2</b><br><b>Weight=15 marks</b><br><b>TL=3</b><br><b>Learning stage = I and D</b> | Demonstrate skills of correct understanding and use of Python data science libraries NumPy and Pandas to prepare and explore data                                    | Demonstrates an insufficient understanding of NumPy and Pandas libraries. Data loading, handling of missing values, and duplicate removal are ineffective. Dates are inaccurately formatted, and the cleaned dataset is erroneous. | Demonstrates a basic understanding of NumPy and Pandas libraries. Data loading, handling of missing values, and duplicate removal need improvement. Dates are inconsistently formatted, and the cleaned dataset contains errors. | Demonstrates a satisfactory understanding of NumPy and Pandas libraries. Data is loaded, but handling of missing values and duplicates could be improved. Dates are somewhat consistent, and the cleaned dataset is mostly accurate. | Demonstrates a proficient understanding of NumPy and Pandas libraries. Data is loaded correctly, missing values are handled adequately, and duplicates are removed. Dates are formatted consistently, and the cleaned dataset is saved correctly. | Demonstrates an exceptional understanding of NumPy and Pandas libraries for data preparation and exploration. Data is loaded correctly, missing values are handled effectively, and duplicates are removed. Dates are consistently formatted, and the cleaned dataset is saved accurately.. |